

Solving a linear system

Question 1 / 10 points

Solve $2x + y = 5$ and $x - y = 1$ using row reduction.

Show the intermediate operations.

Starting matrix

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array} \right]$$

$R1 \leftarrow (1/2) R1$

$$\left[\begin{array}{cc|c} 1 & 1/2 & 5/2 \\ 1 & -1 & 1 \end{array} \right]$$

$R2 \leftarrow R2 + (-1) R1$

$$\left[\begin{array}{cc|c} 1 & 1/2 & 5/2 \\ 0 & -3/2 & -3/2 \end{array} \right]$$

$R2 \leftarrow (-2/3) R2$

$$\left[\begin{array}{cc|c} 1 & 1/2 & 5/2 \\ 0 & 1 & 1 \end{array} \right]$$

$R1 \leftarrow R1 + (-1/2) R2$

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right]$$

Final answer: $x = 2, y = 1$